

Manual Completo de Instalación de VirtualBox, Ubuntu y Hadoop

Introducción

Este manual proporciona una guía detallada para instalar y configurar VirtualBox, una máquina virtual con Ubuntu, y el ecosistema de Hadoop, incluyendo Hadoop, Hive y Spark. Este proceso es ideal para quienes desean aprender y experimentar con tecnologías de Big Data en un entorno local y controlado. Se basa en la información proporcionada, complementada con configuraciones estándar y explicaciones para facilitar el aprendizaje, especialmente para usuarios con conocimientos básicos de sistemas operativos y líneas de comandos.

Hadoop es un marco de código abierto para procesar grandes volúmenes de datos, Hive permite consultas SQL-like sobre datos en Hadoop, y Spark es una plataforma para procesamiento rápido de datos. Este manual configura un cluster de un solo nodo, ideal para pruebas y aprendizaje.

Requisitos

Antes de comenzar, asegúrate de cumplir con los siguientes requisitos:

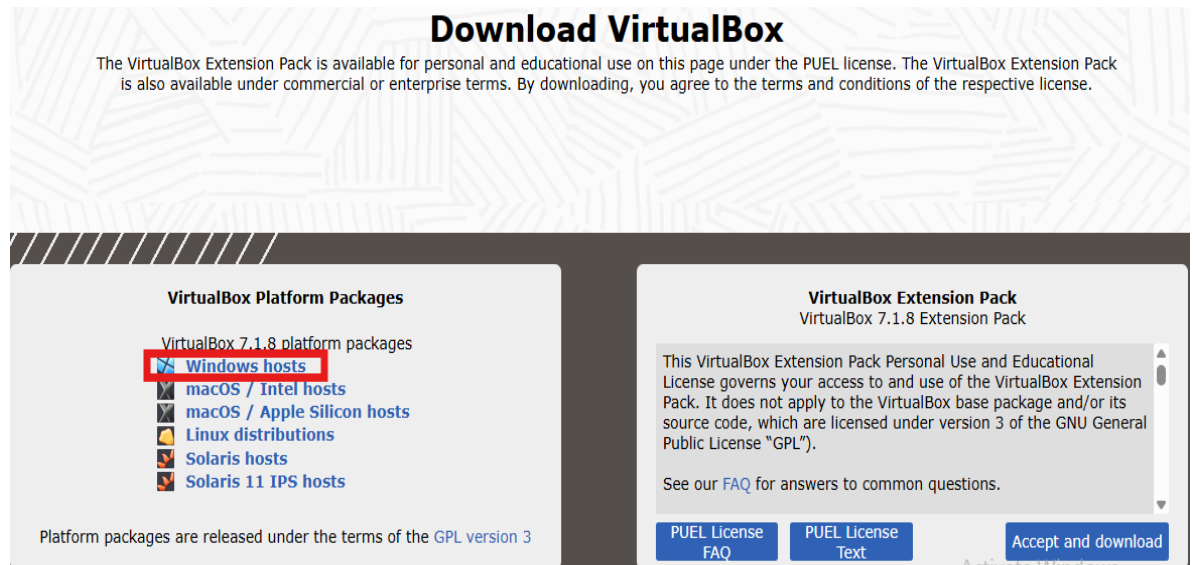
- **Sistema Operativo:** Windows, macOS o Linux compatible con VirtualBox.
- **Acceso a Internet:** Necesario para descargar software.
- **Conocimientos Básicos:** Familiaridad con comandos de terminal y conceptos básicos de sistemas operativos.
- **Espacio en Disco:** Al menos 50 GB para la máquina virtual.
- **Memoria RAM:** Mínimo 4 GB asignados a la máquina virtual para un rendimiento óptimo.

Pasos de Instalación

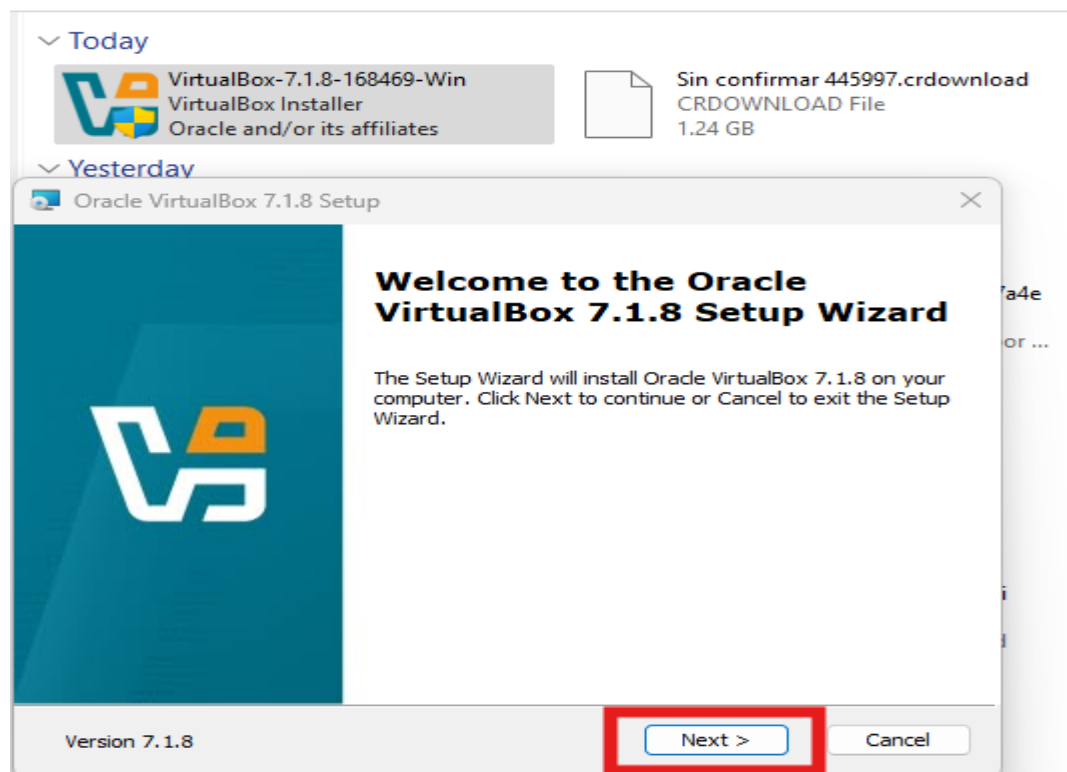
Paso 1: Instalación de VirtualBox

VirtualBox es un hipervisor de código abierto que permite ejecutar sistemas operativos dentro de máquinas virtuales, ideal para crear un entorno aislado para Hadoop.

1. Descarga VirtualBox desde el [sitio oficial de VirtualBox](#).



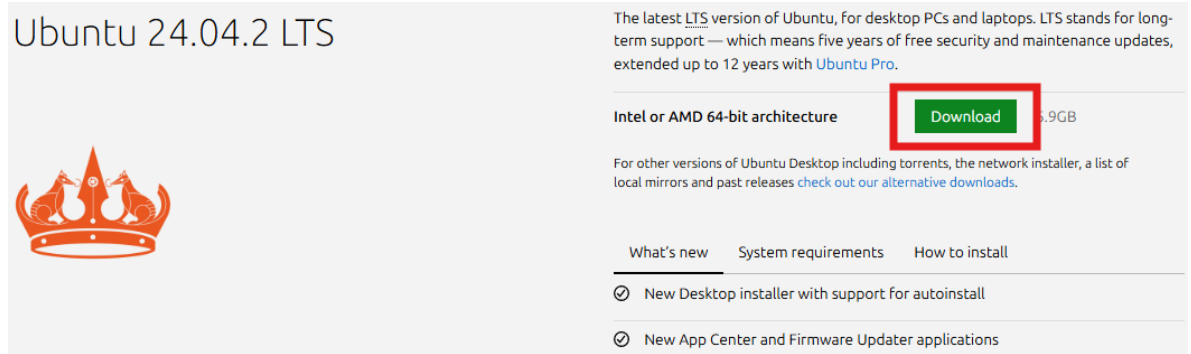
2. Sigue las instrucciones de instalación específicas para tu sistema operativo (Windows, macOS o Linux). (prácticamente presiona “next” siempre)



Paso 2: Descarga de la ISO de Ubuntu

Ubuntu es una distribución de Linux popular, fácil de usar y compatible con Hadoop.

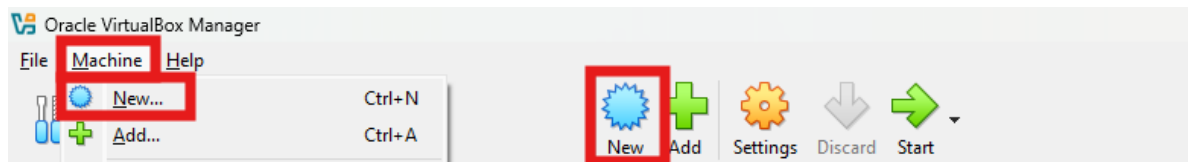
1. Descarga la versión LTS de Ubuntu desde el [sitio oficial de Ubuntu](#).



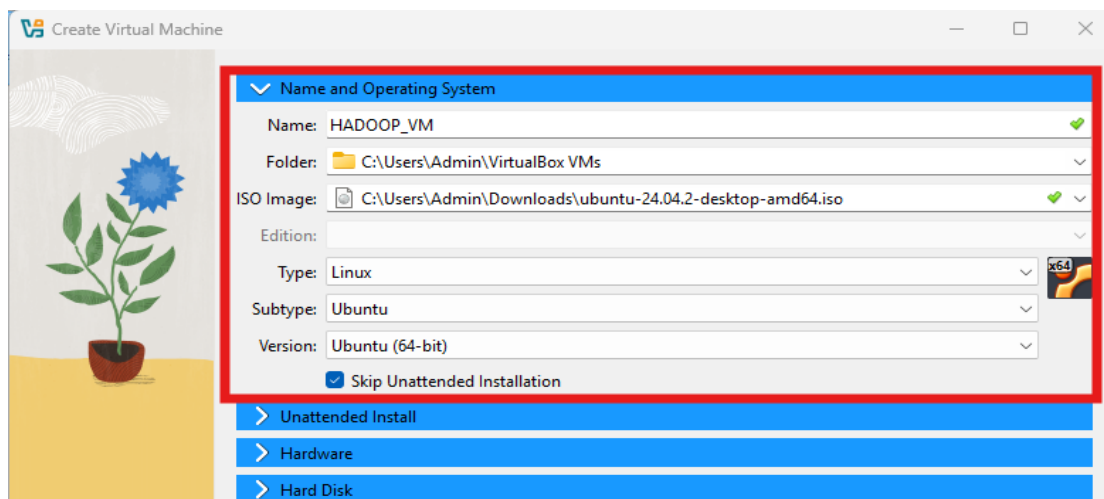
2. Guarda el archivo ISO en una ubicación accesible en tu computadora.

Paso 3: Configuración de VirtualBox

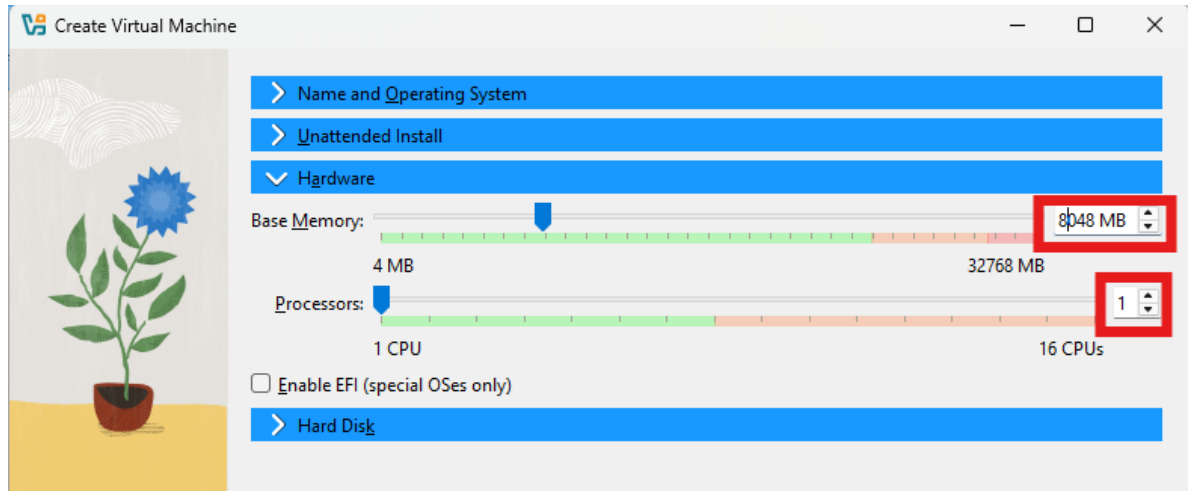
1. Abre VirtualBox y crea una nueva máquina virtual:



- **Nombre:** "HADOOP_VM"
- **Tipo:** Linux
- **Versión:** Ubuntu (64-bit)
- Selecciona la ISO de Ubuntu descargada.

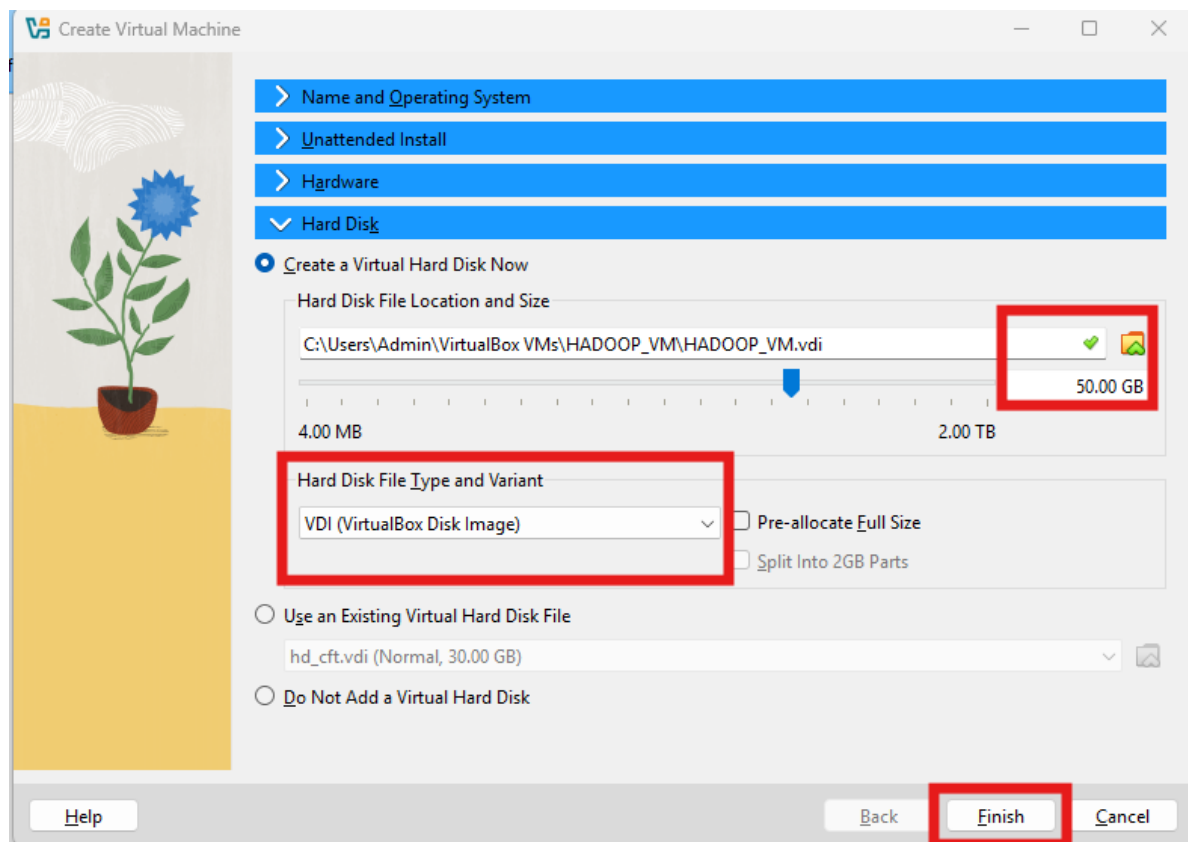


2. Asigna memoria RAM (recomendado al menos 4 GB para un rendimiento óptimo).



3. Configura el almacenamiento:

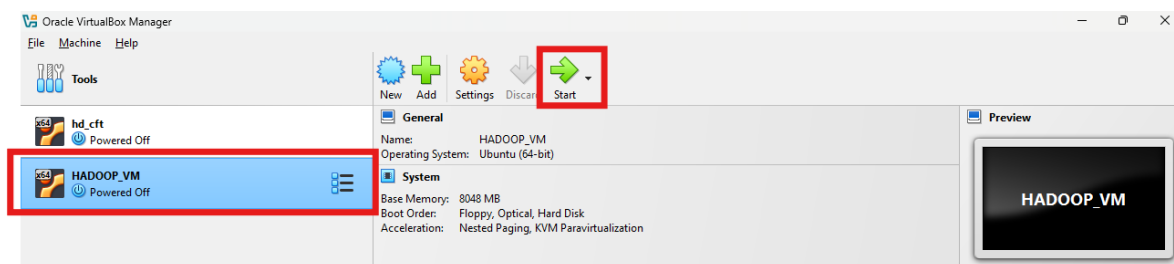
- Crea un disco virtual dinámico con al menos 50 GB.



4. Usa el modo experto si está disponible y habilita la “Unattended Install” para simplificar el proceso.

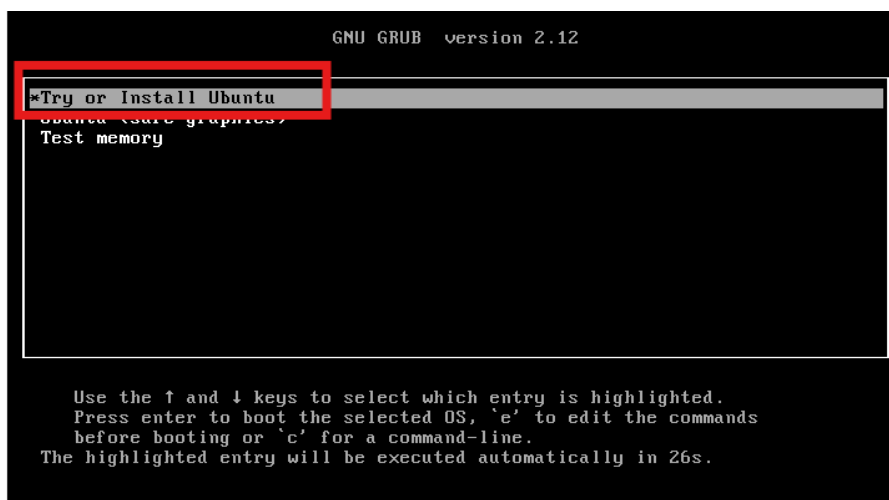
Paso 4: Instalación de Ubuntu

1. Inicia la máquina virtual para comenzar la instalación de Ubuntu.

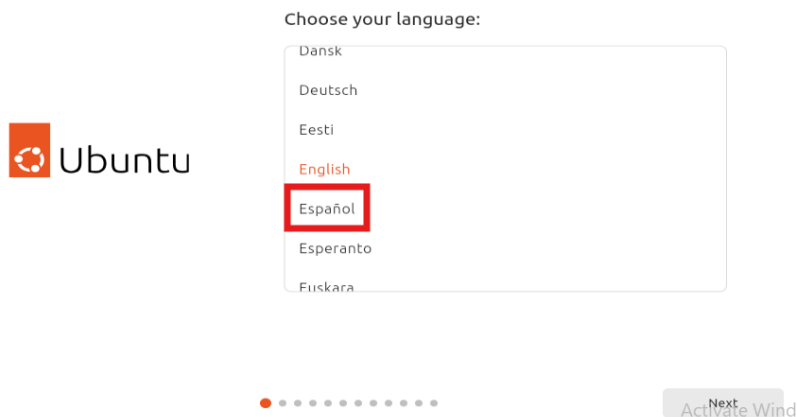


2. Sigue los pasos del instalador:

- “Try or Install Ubuntu”



- Selecciona el idioma deseado (por ejemplo, español).



- Accesibilidad a Ubuntu

Accessibility in Ubuntu

Customise Ubuntu to your needs before you set up. You can change them later in System Settings.

👁 Seeing	>
👂 Hearing	>
🖱 Typing	>
🖱 Pointing and clicking	>
🔍 Zoom	>



Next

- Configura el teclado según tu preferencia.

Keyboard layout



Select your keyboard layout

Detect

English (South Africa)

English (UK)

English (US)

Esperanto

Estonian

Select your keyboard variant:

English (US)



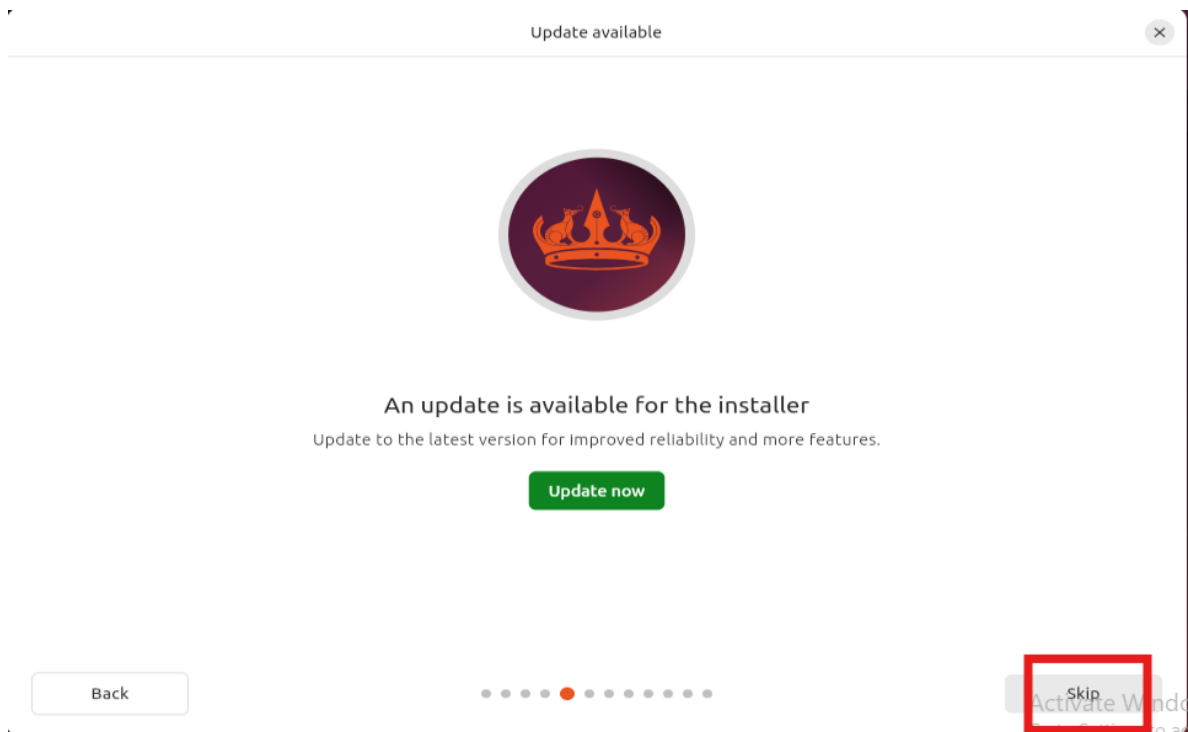
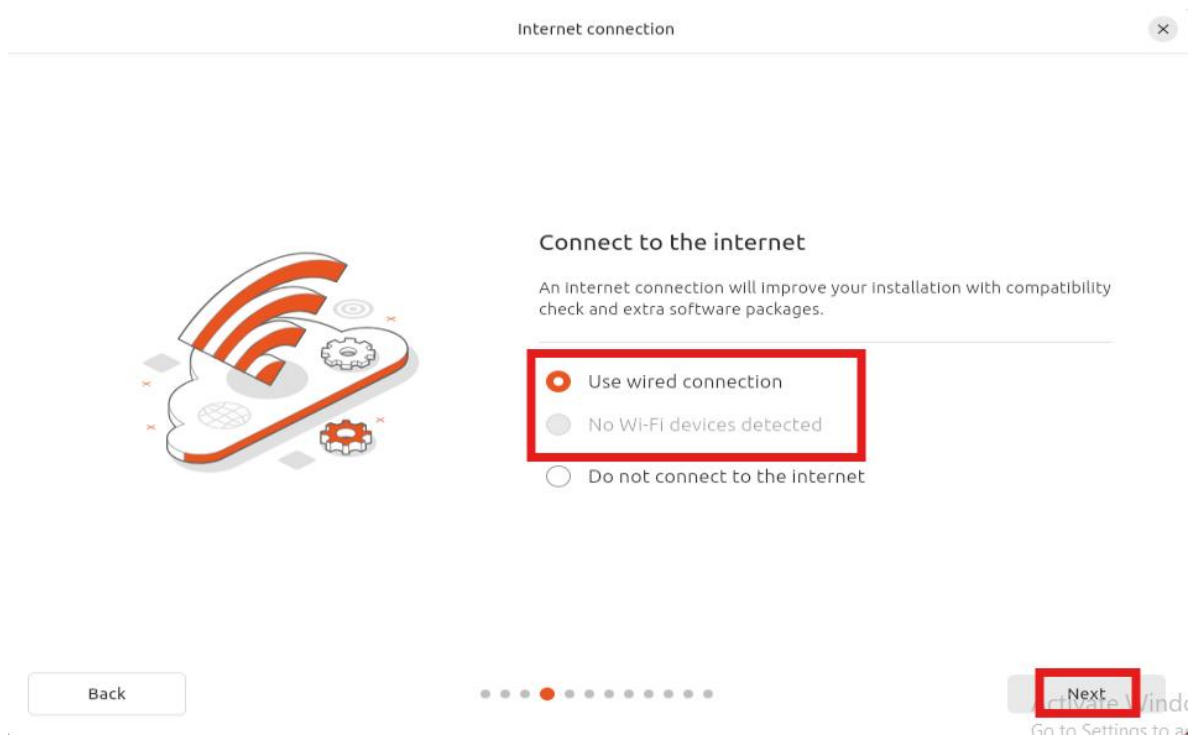
Type here to test your keyboard

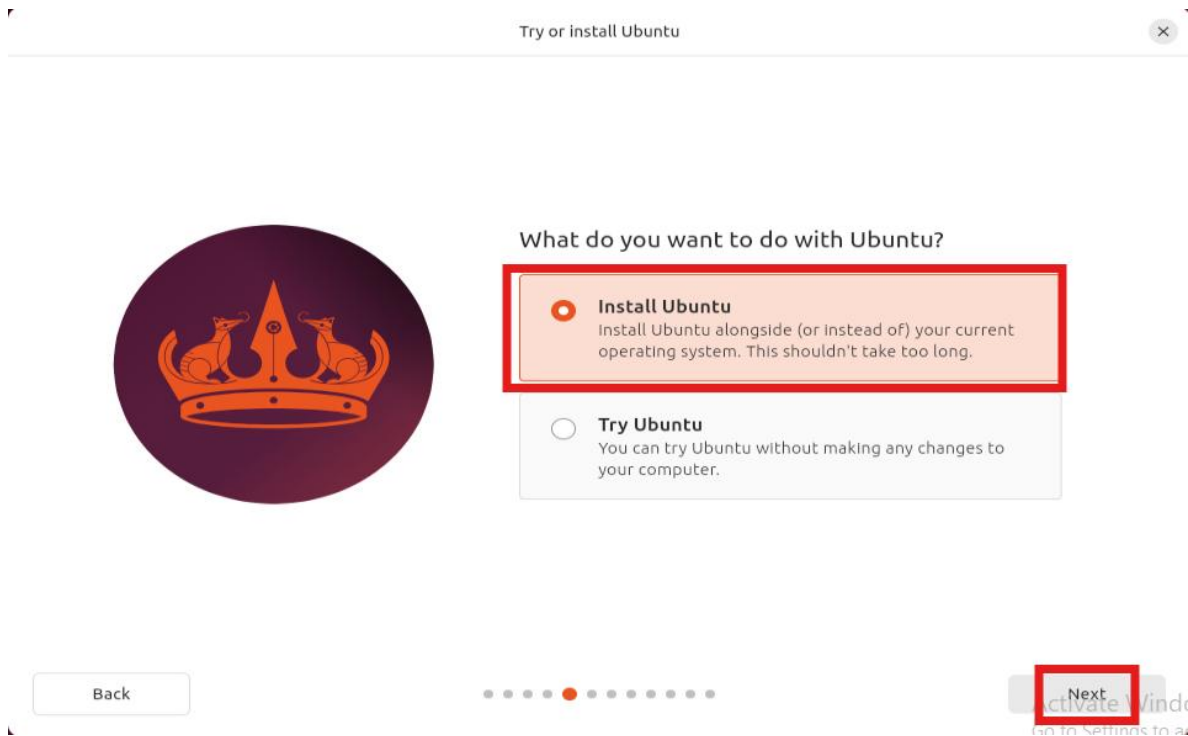
Back



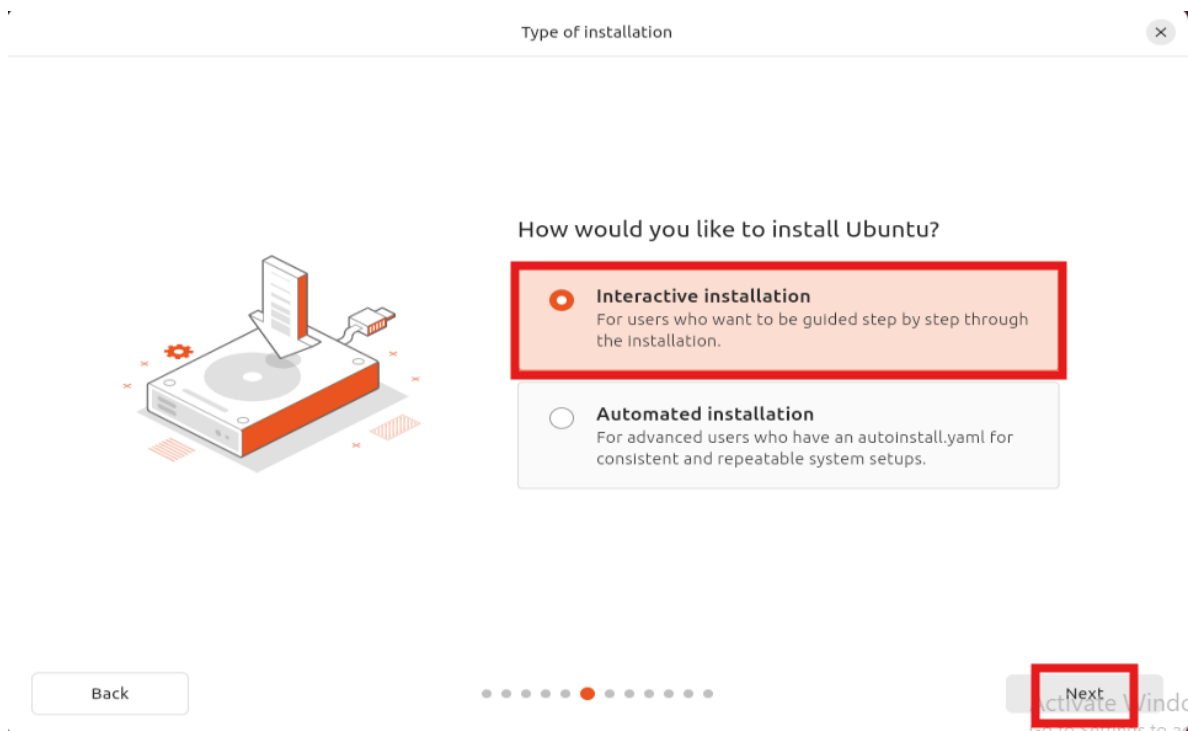
Next

- Asegúrate de tener una conexión a internet (preferiblemente por cable).





- Elige la opción de instalación interactiva.





What apps would you like to install to start with?

**Default selection**

Just the essentials, web browser and basic utilities.

**Extended selection**

An offline-friendly selection of office tools, utilities and web browser.

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Activate Windows
Go to Settings to activate Windows.

- Selecciona "Next".

**Install recommended proprietary software?**

Ubuntu ships with no proprietary software by default. Installing additional software may improve your computer's performance.



Install third-party software for graphics and Wi-Fi hardware

Including but not limited to NVIDIA drivers and similar



Download and install support for additional media formats

Including but not limited to MP3, MP4, MOV and similar

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Activate Windows
Go to Settings to activate Windows.

- Confirma borra el disco cuando lo solicite

Disk setup



How do you want to install Ubuntu?

☒ Erase disk and install Ubuntu
Start from scratch on your selected disk.

Advanced features... None selected

☐ Manual installation
For advanced users seeking customized disk setups.

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Next

- Crea un usuario:
 - **Nombre de usuario:** HADOOP
 - **Contraseña:** 123 (o una de tu elección, pero sigue el guía para consistencia).

Create your account



Create your account

Your name
HADDOP ✓

Your computer's name
hadoop-VirtualBox ✓

Your username
hadoop ✓

Password
123 Hide Weak password

Confirm password
123 ✓

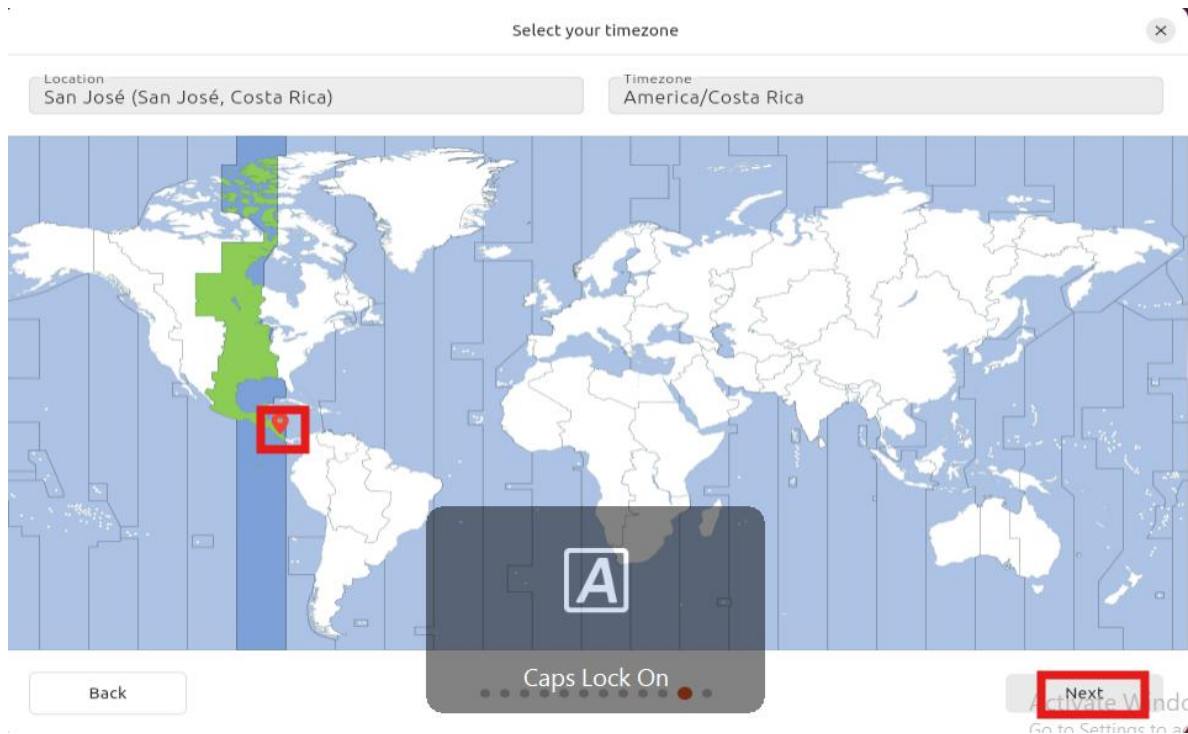
☒ Require my password to log in

☐ Use Active Directory

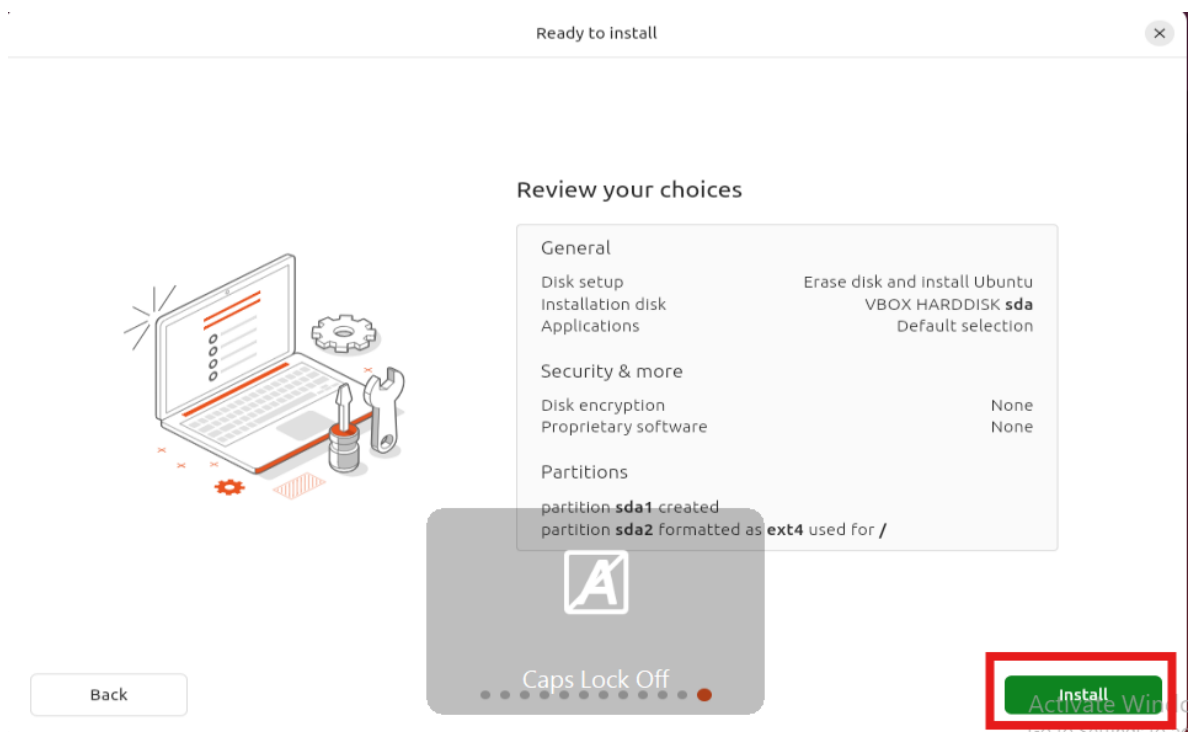
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Next

- Configura la zona horaria (por ejemplo, "Mucho Uso Horario CR" para Costa Rica).



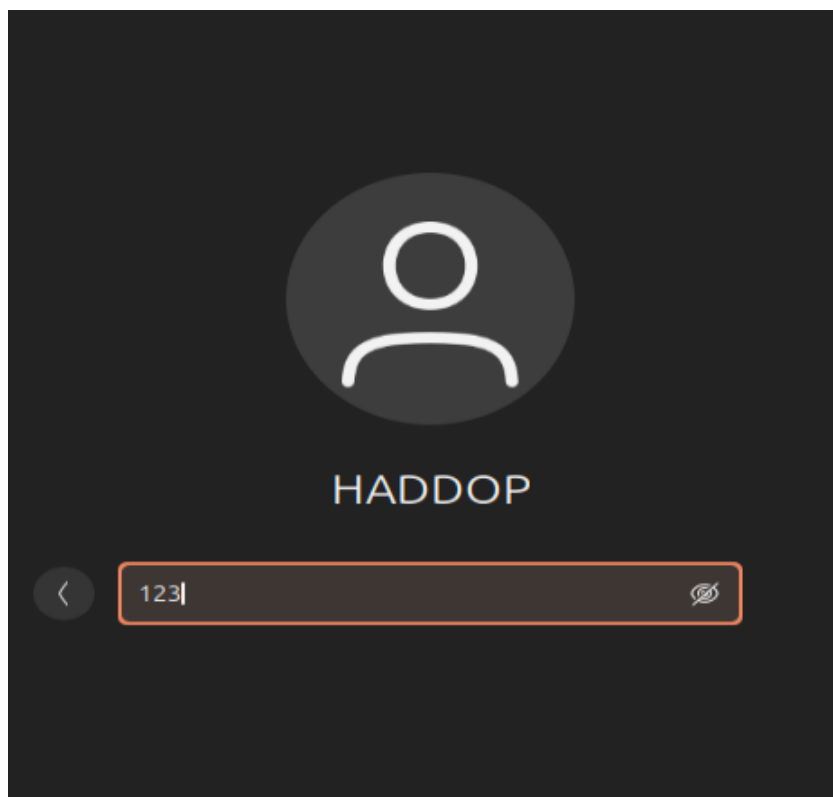
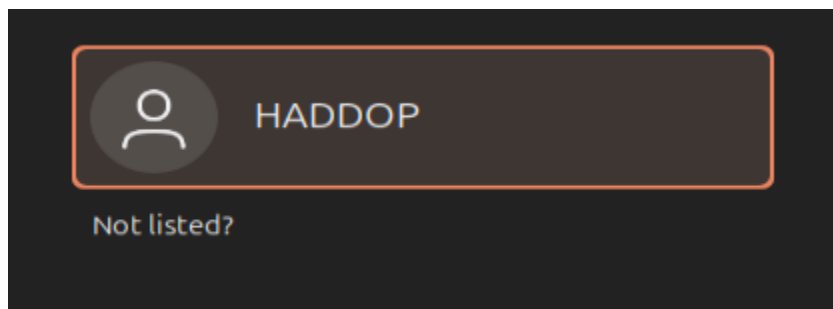
3. Finaliza la instalación haciendo clic en "Instalar".



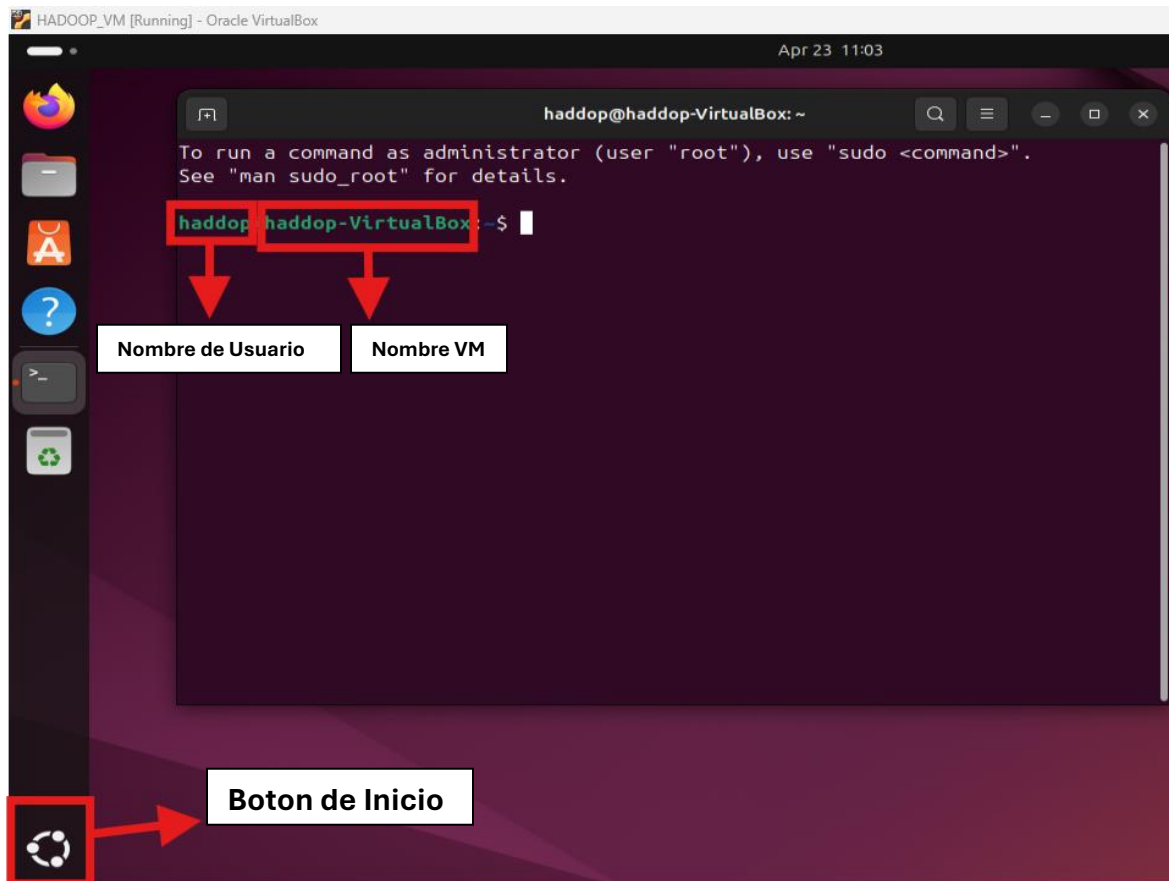
Paso 5: Instalación de Java

Java es un requisito fundamental para ejecutar Hadoop, ya que sus componentes están escritos en este lenguaje.

1. Ingresamos a la VM



1. Abre una terminal en la máquina virtual de Ubuntu. (presiona Ctrl + Alt + T)



2. Actualiza los repositorios de paquetes: **sudo apt update**,

```
haddop@haddop-VirtualBox: ~$ sudo apt update
[sudo] password for haddop:
Hit:1 http://cr.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://cr.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://cr.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
101 packages can be upgraded. Run 'apt list --upgradable' to see them.
haddop@haddop-VirtualBox: ~$
```

3. contraseña: 123, “Enter”

4. Instala OpenJDK 11: **sudo apt install openjdk-11-jdk -y**

```
hadoop@hadoop-VirtualBox: ~  
101 packages can be upgraded. Run 'apt list --upgradable' to see them.  
hadoop@hadoop-VirtualBox:~$ sudo apt install openjdk-11-jdk -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java  
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev  
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev  
  openjdk-11-jdk-headless openjdk-11-jre openjdk-11-jre-headless x11proto-dev  
  xorg-sgml-doctools xtrans-dev  
Suggested packages:  
  default-jre libice-doc libsm-doc libx11-doc libxcb-doc libxt-doc  
  openjdk-11-demo openjdk-11-source visualvm fonts-ipafont-gothic  
  fonts-ipafont-mincho fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic  
The following NEW packages will be installed:  
  ca-certificates-java fonts-dejavu-extra java-common libatk-wrapper-java  
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev  
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev openjdk-11-jdk  
  openjdk-11-jdk-headless openjdk-11-jre openjdk-11-jre-headless x11proto-dev  
  xorg-sgml-doctools xtrans-dev  
0 upgraded, 20 newly installed, 0 to remove and 101 not upgraded.  
Need to get 122 MB of archives.  
After this operation, 275 MB of additional disk space will be used.
```

5. Verifica la instalación:

- **java -version**
- **javac -version**
- Deberías ver una salida indicando OpenJDK 11.

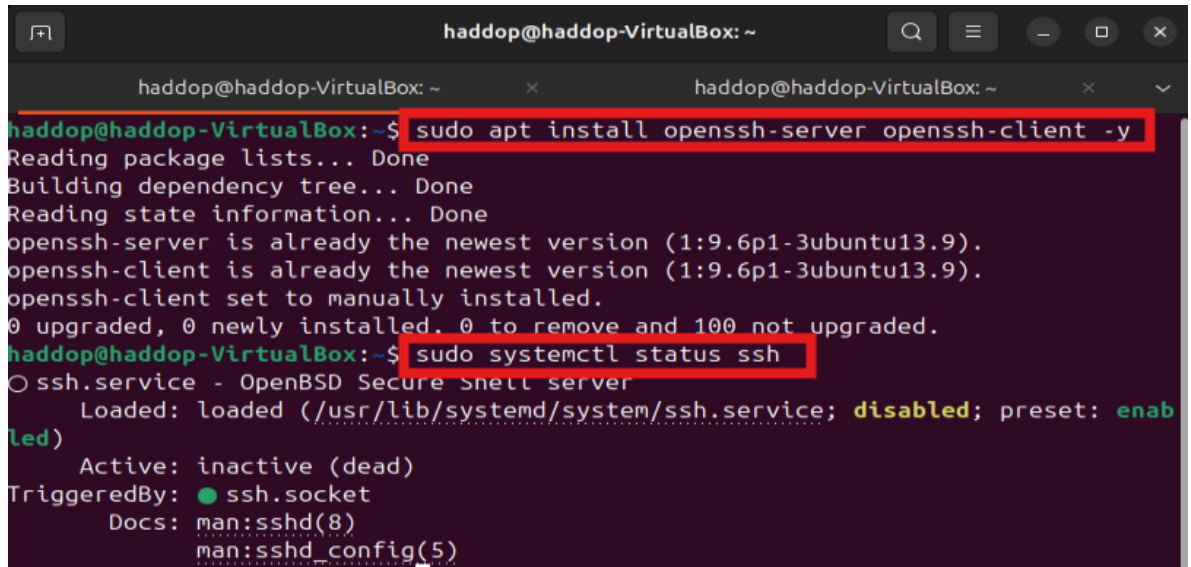
```
hadoop@hadoop-VirtualBox: ~  
hadoop@hadoop-VirtualBox:~$ java -version; javac -version  
openjdk version "11.0.26" 2025-01-21  
OpenJDK Runtime Environment (build 11.0.26+4-post-Ubuntu-1ubuntu124.04)  
OpenJDK 64-Bit Server VM (build 11.0.26+4-post-Ubuntu-1ubuntu124.04, mixed mode,  
  sharing)  
javac 11.0.26  
hadoop@hadoop-VirtualBox:~$
```

Nota: El guía original instalaba OpenJDK 8 pero configuraba JAVA_HOME para Java 11. Para consistencia, este manual usa Java 11, que es compatible con Hadoop 3.4.1 según la [documentación de Hadoop](#).

Paso 6: Instalación de SSH

SSH (Secure Shell) permite la comunicación segura entre procesos de Hadoop, incluso en un cluster de un solo nodo.

1. Instala el servidor y cliente SSH: **sudo apt install openssh-server openssh-client -y**
2. Verifica que el servicio SSH esté activo: **sudo systemctl status ssh**

A terminal window titled 'hadoop@hadoop-VirtualBox: ~' showing the execution of two commands. The first command, 'sudo apt install openssh-server openssh-client -y', is highlighted with a red box and shows the output of the package manager, indicating that the software is already up-to-date. The second command, 'sudo systemctl status ssh', is also highlighted with a red box and shows the status of the SSH service, which is currently 'disabled' but 'enabled' by default.

```
hadoop@hadoop-VirtualBox: ~  
hadoop@hadoop-VirtualBox: ~  
hadoop@hadoop-VirtualBox:~$ sudo apt install openssh-server openssh-client -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
openssh-server is already the newest version (1:9.6p1-3ubuntu13.9).  
openssh-client is already the newest version (1:9.6p1-3ubuntu13.9).  
openssh-client set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 100 not upgraded.  
hadoop@hadoop-VirtualBox:~$ sudo systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)  
   Active: inactive (dead)  
TriggeredBy: ● ssh.socket  
   Docs: man:sshd(8)  
         man:sshd_config(5)
```

Paso 7: Creación de Usuario Hadoop

Hadoop se ejecuta mejor bajo un usuario dedicado para aislar sus procesos.

1. Crea un nuevo usuario: **sudo adduser hdoop**
2. Cambia al usuario creado: **su - hdoop**

```
hadoop@hadoop-VirtualBox: /home/hadoop
hadoop@hadoop-VirtualBox:~$ sudo adduser hadoop
info: Adding user 'hadoop' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'hadoop' (1001) ...
info: Adding new user 'hadoop' (1001) with group 'hadoop (1001)' ...
info: Creating home directory '/home/hadoop' ...
info: Copying files from '/etc/skel' ...
New password: 123
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for hadoop
Enter the new value, or press ENTER for the default
  Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user 'hadoop' to supplemental / extra groups 'users' ...
info: Adding user 'hadoop' to group 'users' ...
hadoop@hadoop-VirtualBox:~$ su hadoop
Password:
hadoop@hadoop-VirtualBox: /home/hadoop$ cd
```

Presiona "Enter" Siempre

Con "cd" regresamos a Home

Paso 8: Configuración de SSH para el Usuario Hadoop

Hadoop requiere SSH sin contraseña para comunicarse consigo mismo en un cluster de un solo nodo.

1. Genera una clave SSH: **ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa**
2. Agrega la clave pública a las claves autorizadas: **cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys**
3. Establece permisos correctos: **chmod 0600 ~/.ssh/authorized_keys**
4. Prueba la conexión SSH: **ssh localhost**
 - Deberías conectarte sin necesidad de contraseña.

```
hadoop@hadoop-VirtualBox:~$ ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
Generating public/private rsa key pair.
Your identification has been saved in /home/hadoop/.ssh/id_rsa
Your public key has been saved in /home/hadoop/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:46FzK/QQ9Bs6PiHiBq02d92bItIYcDXo0RtrpYzNFCE hadoop@hadoop-VirtualBox
The key's randomart image is:
+---[RSA 3072]-----+
|      Eooo          |
|      o.*          |
|      . 0 B         |
|      + 0           |
|      + S           |
|      ....* o       |
|      ..*.X .       |
|      ..o..=0 O.     |
|      |o+=...o**o.   |
|      +----[SHA256]-----+
hadoop@hadoop-VirtualBox:~$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
hadoop@hadoop-VirtualBox:~$ chmod 0600 ~/.ssh/authorized_keys
hadoop@hadoop-VirtualBox:~$ ssh localhost
hadoop@localhost's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.11.0-24-generic x86_64)
```

~/.ssh/id_rsa.pub

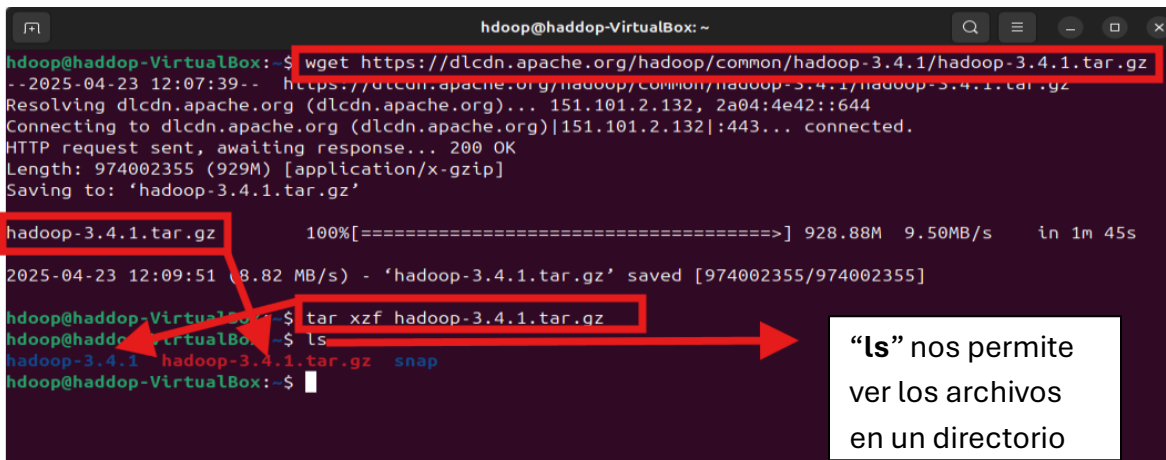
Paso 9: Descarga e Instalación de Hadoop

Hadoop es el núcleo del ecosistema, proporcionando almacenamiento (HDFS) y procesamiento (YARN, MapReduce).

1. Descarga Hadoop 3.4.1:

wget https://dlcdn.apache.org/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz

2. Extrae el archivo: **tar xzf hadoop-3.4.1.tar.gz**

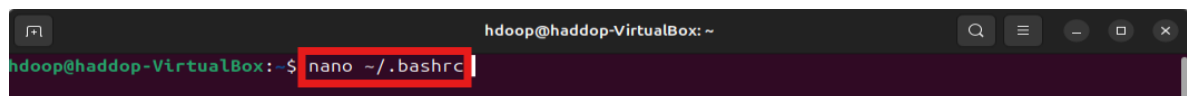


```
hadoop@hadoop-VirtualBox: ~  
$ wget https://dlcdn.apache.org/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz  
--2025-04-23 12:07:39-- https://dlcdn.apache.org/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz  
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644  
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 974002355 (929M) [application/x-gzip]  
Saving to: 'hadoop-3.4.1.tar.gz'  
  
hadoop-3.4.1.tar.gz      100%[=====] 928.88M  9.50MB/s   in 1m 45s  
2025-04-23 12:09:51 (8.82 MB/s) - 'hadoop-3.4.1.tar.gz' saved [974002355/974002355]  
  
hadoop@hadoop-VirtualBox: ~$ tar xzf hadoop-3.4.1.tar.gz  
hadoop@hadoop-VirtualBox: ~$ ls  
hadoop-3.4.1  hadoop-3.4.1.tar.gz  snap  
hadoop@hadoop-VirtualBox: ~$
```

“ls” nos permite ver los archivos en un directorio

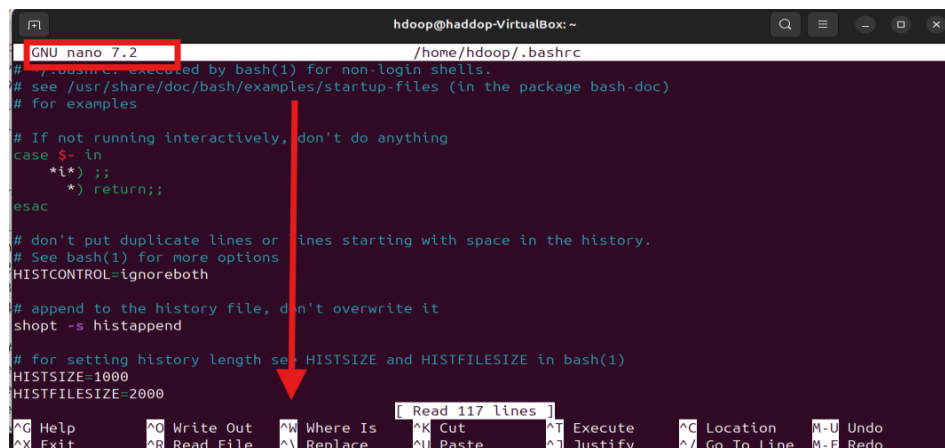
3. Configura las variables de entorno:

- Abre el archivo. bashrc: **nano ~/.bashrc**



```
hadoop@hadoop-VirtualBox: ~  
$ nano ~/.bashrc
```

- Agrega las siguientes líneas al final:



```
GNU nano 7.2 /home/hadoop/.bashrc  
# .bashrc: executed by bash(1) for non-login shells.  
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)  
# for examples  
  
# If not running interactively, don't do anything  
case $- in  
  *(i*) ;;  
  *) return;;  
esac  
  
# don't put duplicate lines or lines starting with space in the history.  
# See bash(1) for more options  
HISTCONTROL=ignoreboth  
  
# append to the history file, don't overwrite it  
shopt -s histappend  
  
# for setting history length see HISTSIZE and HISTFILESIZE in bash(1)  
HISTSIZE=1000  
HISTFILESIZE=2000
```

- # Configuración de Hadoop
- export HADOOP_HOME=/home/hadoop/hadoop-3.4.1
- export HADOOP_INSTALL=\$HADOOP_HOME
- export HADOOP_MAPRED_HOME=\$HADOOP_HOME
- export HADOOP_COMMON_HOME=\$HADOOP_HOME
- export HADOOP_HDFS_HOME=\$HADOOP_HOME
- export YARN_HOME=\$HADOOP_HOME
- export
HADOOP_COMMON_LIB_NATIVE_DIR=\$HADOOP_HOME/lib/native
- export PATH=\$PATH:\$HADOOP_HOME/bin:\$HADOOP_HOME/sbin
- export HADOOP_OPTS="-
Djava.library.path=\$HADOOP_HOME/lib/native"
- export HADOOP_CONF_DIR=\$HADOOP_HOME/etc/Hadoop

- # Configuración de Hive
- export HIVE_HOME=/home/hadoop/apache-hive-4.0.1-bin
- export PATH=\$PATH:\$HIVE_HOME/bin

- # Configuración de Spark (versión 3.5.5)
- export SPARK_HOME=/home/hadoop/spark-3.5.5-bin-hadoop3
- export PATH=\$PATH:\$SPARK_HOME/bin
- export PYTHONPATH=\$SPARK_HOME/python:\$PYTHONPATH

- # Variables de Java (opcional, si es necesario)
- export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
- export PATH=\$PATH:\$JAVA_HOME/bin

```
hadoop@hadoop-VirtualBox: ~  
GNU nano 7.2 /home/hadoop/.bashrc *  
. /etc/bash_completion  
fi  
fi  
  
# Configuración de Hadoop  
export HADOOP_HOME=/home/hadoop/hadoop-3.4.1  
export HADOOP_INSTALL=$HADOOP_HOME  
export HADOOP_MAPRED_HOME=$HADOOP_HOME  
export HADOOP_COMMON_HOME=$HADOOP_HOME  
export HADOOP_HDFS_HOME=$HADOOP_HOME  
export YARN_HOME=$HADOOP_HOME  
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native  
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin  
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"  
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop  
  
# Configuración de Hive  
export HIVE_HOME=/home/hadoop/apache-hive-4.0.1-bin  
export PATH=$PATH:$HIVE_HOME/bin  
  
# Configuración de Spark (versión 3.5.5)  
export SPARK_HOME=/home/hadoop/spark-3.5.5-bin-hadoop3  
export PATH=$PATH:$SPARK_HOME/bin  
export PYTHONPATH=$SPARK_HOME/python:$PYTHONPATH  
  
# Variables de Java (opcional, si es necesario)  
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64  
export PATH=$PATH:$JAVA_HOME/bin  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

- Guarda y cierra el archivo.

Presiona "Ctrl + X" para salir

Al presionar “Ctrl + X” aparecerá “Save modified buffer?” escribe “Y”

```
GNU nano 7.2 /home/hadoop/
export HADOOP_INSTALL=$HADOOP_HOME
export HADOOP_MAPRED_HOME=$HADOOP_HOME
export HADOOP_COMMON_HOME=$HADOOP_HOME
export HADOOP_HDFS_HOME=$HADOOP_HOME
export YARN_HOME=$HADOOP_HOME
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib/native"
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop

# Configuración de Hive
export HIVE_HOME=/home/hadoop/apache-hive-4.0.1-bin
export PATH=$PATH:$HIVE_HOME/bin

# Configuración de Spark (versión 3.5.5)
export SPARK_HOME=/home/hadoop/spark-3.5.5-bin-hadoop3
export PATH=$PATH:$SPARK_HOME/bin
export PYTHONPATH=$SPARK_HOME/python:$PYTHONPATH

# Variables de Java (opcional, si es necesario)
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$PATH:$JAVA_HOME/bin

Save modified buffer?
Y Yes
N NO ^C Cancel
```

luego “Enter”

```
File Name to Write: /home/hadoop/.bashrc
^G Help M-D DOS Format M-A Append
^C Cancel M-M Mac Format M-P Prepend
```

4. Aplica los cambios: **source ~/.bashrc**

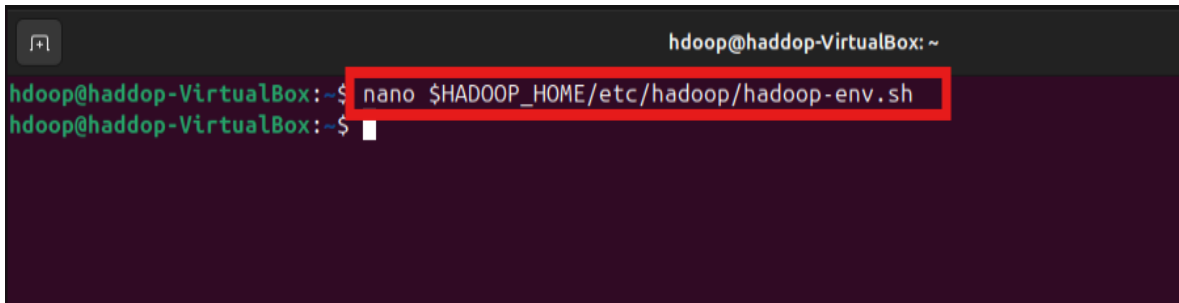
```
hadoop@hadoop-VirtualBox: ~
hadoop@hadoop-VirtualBox:~$ nano ~/.bashrc
hadoop@hadoop-VirtualBox:~$ source ~/.bashrc
hadoop@hadoop-VirtualBox:~$
```

Paso 10: Configuración de Hadoop

Hadoop requiere configuraciones específicas para operar en modo pseudo-distribuido.

1. Edita el archivo de entorno de Hadoop:

nano \$HADOOP_HOME/etc/hadoop/hadoop-env.sh

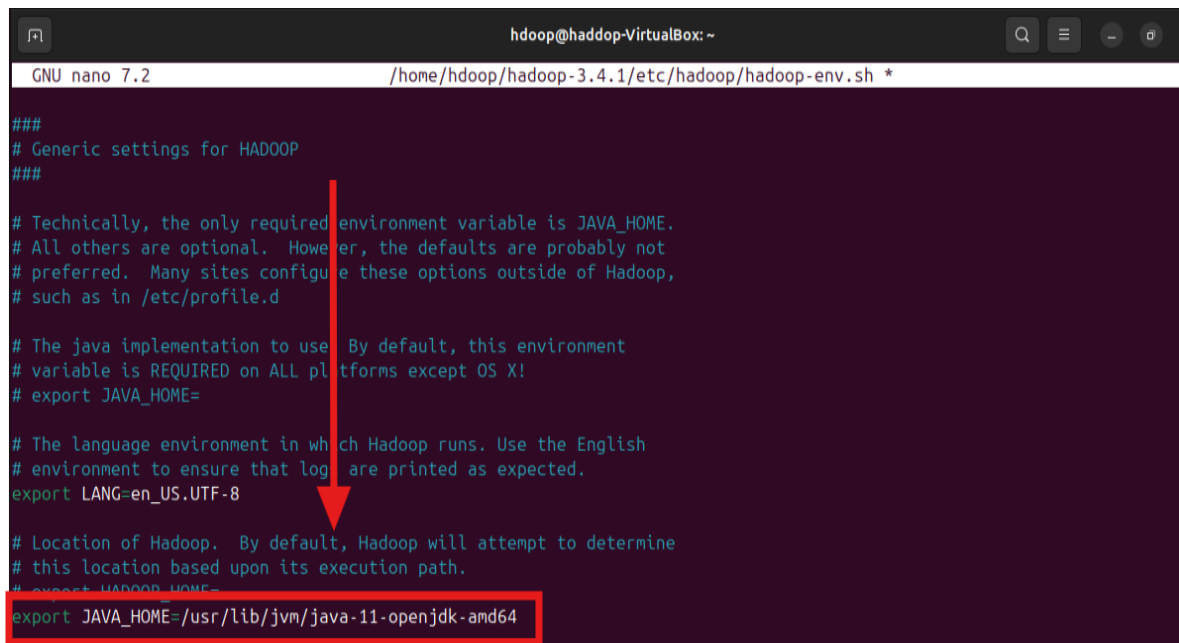


A terminal window titled 'hdoop@hadoop-VirtualBox: ~' shows the command 'nano \$HADOOP_HOME/etc/hadoop/hadoop-env.sh' being entered at the prompt. The command is highlighted with a red box.

Al ingresar al documento `hadoop-env.sh` baja hasta encontrar

- Asegúrate de que incluya:

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64



A terminal window showing the contents of the file `/home/hdoop/hadoop-3.4.1/etc/hadoop/hadoop-env.sh` in nano 7.2. The file contains several comments and environment variable exports. A red arrow points to the line `export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64`, which is highlighted with a red box.

Al presionar “Ctrl + X” aparecerá “Save modified buffer?” escribe “Y”, además por último “Enter”.

2. Configura core-site.xml: **nano \$HADOOP_HOME/etc/hadoop/core-site.xml**

- Agrega:

```
<configuration>
```

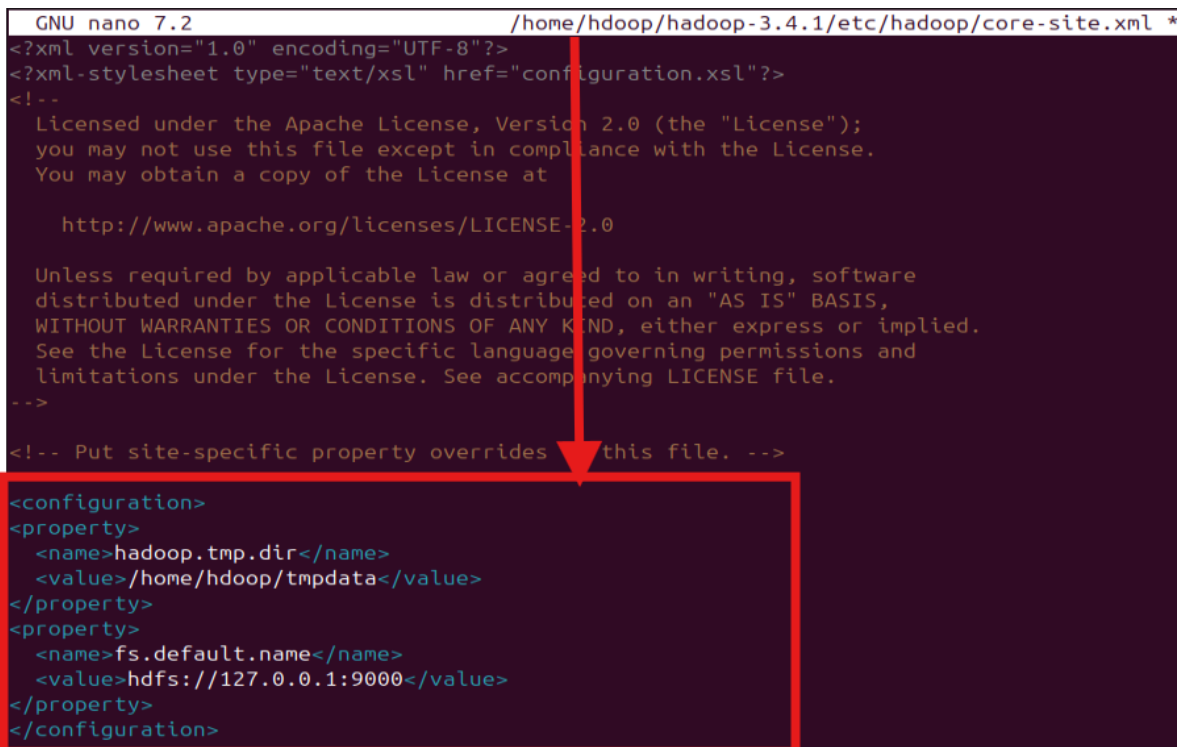
```
<property>
```

```
<name>fs.defaultFS</name>
```

```
<value>hdfs://localhost:9000</value>
```

```
</property>
```

```
</configuration>
```



```
GNU nano 7.2 /home/hadoop/hadoop-3.4.1/etc/hadoop/core-site.xml *
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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  you may not use this file except in compliance with the License.
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    http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->
<configuration>
<property>
  <name>hadoop.tmp.dir</name>
  <value>/home/hadoop/tmpdata</value>
</property>
<property>
  <name>fs.default.name</name>
  <value>hdfs://127.0.0.1:9000</value>
</property>
</configuration>
```

Al presionar “Ctrl + X” aparecerá “Save modified buffer?” escribe “Y”, además por último “Enter”.

3. Configura hdfs-site.xml: **nano \$HADOOP_HOME/etc/hadoop/hdfs-site.xml**

- Agrega:

```
<configuration>

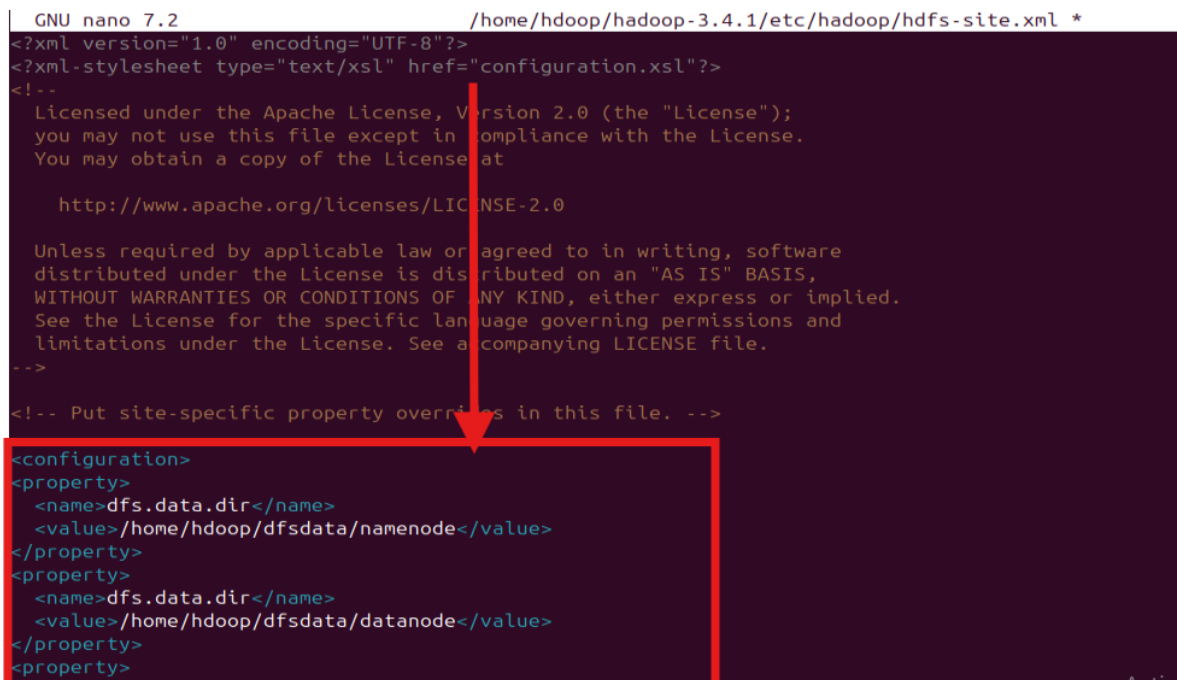
  <property>

    <name>dfs.replication</name>

    <value>1</value>

  </property>

</configuration>
```



```
GNU nano 7.2 /home/hadoop/hadoop-3.4.1/etc/hadoop/hdfs-site.xml *
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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 distributed under the License is distributed on an "AS IS" BASIS,
 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
 See the License for the specific language governing permissions and
 limitations under the License. See a accompanying LICENSE file.
-->
<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>dfs.data.dir</name>
  <value>/home/hadoop/dfsdata/namenode</value>
</property>
<property>
  <name>dfs.data.dir</name>
  <value>/home/hadoop/dfsdata/datanode</value>
</property>
<property>
```

Al presionar “Ctrl + X” aparecerá “Save modified buffer?” escribe “Y”, además por último “Enter”.

4. Configura mapred-site.xml:

nano \$HADOOP_HOME/etc/hadoop/mapred-site.xml

Agrega:

```
<configuration>

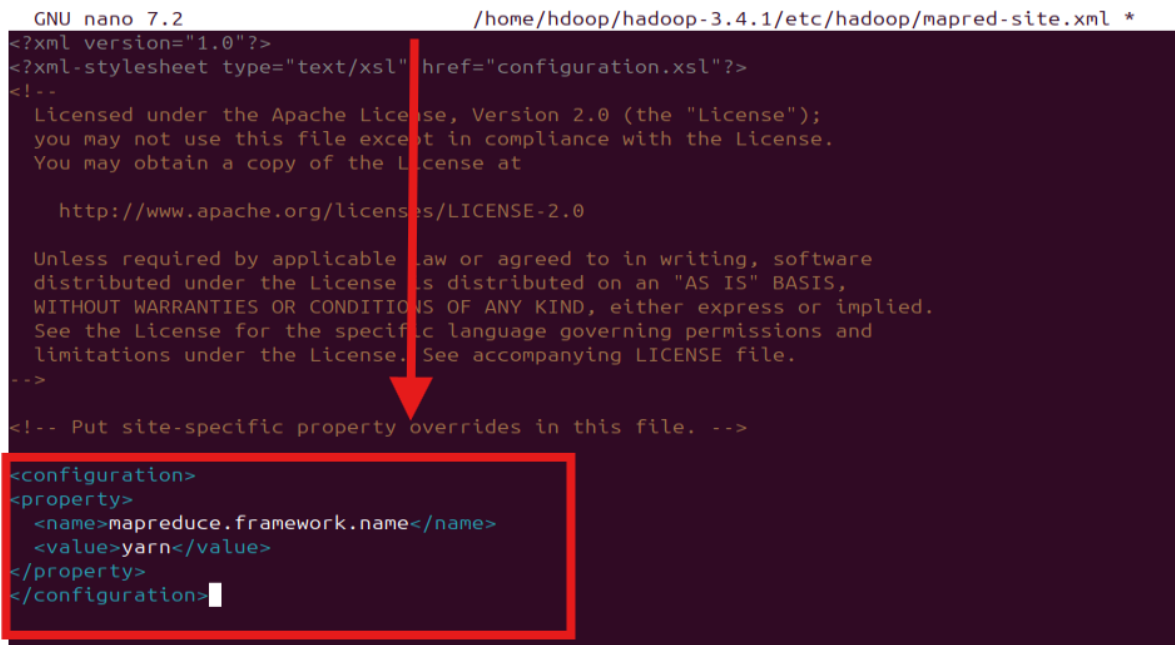
  <property>

    <name>mapreduce.framework.name</name>

    <value>yarn</value>

  </property>

</configuration>
```



```
GNU nano 7.2 /home/hadoop/hadoop-3.4.1/etc/hadoop/mapred-site.xml *
<?xml version="1.0"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
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    http://www.apache.org/licenses/LICENSE-2.0

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 distributed under the License is distributed on an "AS IS" BASIS,
 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
 See the License for the specific language governing permissions and
 limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
<property>
  <name>mapreduce.framework.name</name>
  <value>yarn</value>
</property>
</configuration>
```

Al presionar “Ctrl + X” aparecerá “Save modified buffer?” escribe “Y”, además por último “Enter”.

5. Configura yarn-site.xml: **nano \$HADOOP_HOME/etc/hadoop/yarn-site.xml**

- Agrega:

```
<configuration>
```

```
<!--configuracion de YARN -->
```

```
<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>
<property>
  <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
<property>
  <name>yarn.resourcemanager.hostname</name>
  <value>127.0.0.1</value>
</property>
<property>
  <name>yarn.acl.enable</name>
  <value>0</value>
</property>
<property>
  <name>yarn.nodemanager.env-whitelist</name>
  <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP_MAPRED_HOME</value>
</property>
```

```
<!--configuracion de HIVE -->
```

```
<property>
```

```
<name>hadoop.proxyuser.db_user.groups</name>
```

```
<value>*</value>
```

```
</property>
```

```
<property>
```

```
<name>hadoop.proxyuser.db_user.hosts</name>
```

```
<value>*</value>
```

```
</property>
```

```
<property>
```

```
<name>hadoop.proxyuser.server.hosts</name>
```

```

<value>*</value>

</property>

<property>

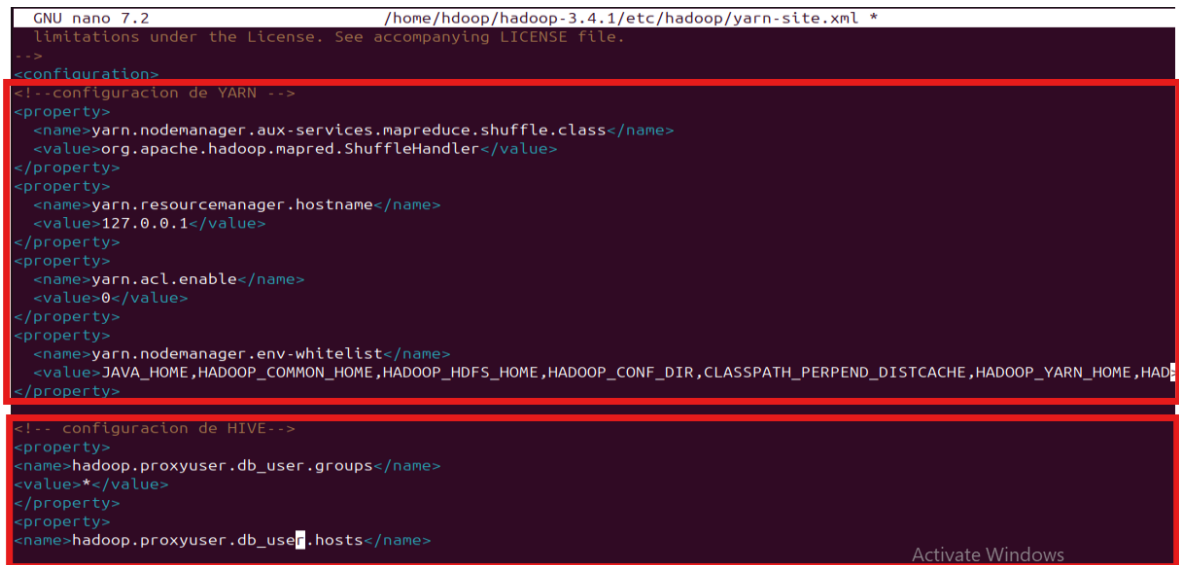
<name>hadoop.proxyuser.server.groups</name>

<value>*</value>

</property>

</configuration>

```



```

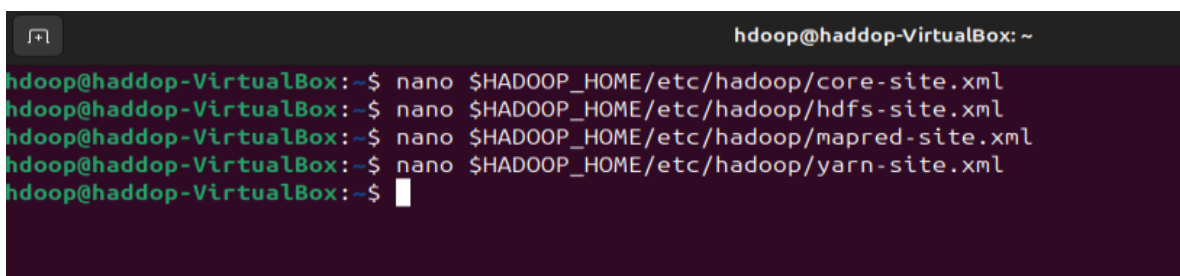
GNU nano 7.2 /home/hdoop/hadoop-3.4.1/etc/hadoop/yarn-site.xml *
limitations under the License. See accompanying LICENSE file.
-->
<configuration>
<!-- configuracion de YARN -->
<property>
  <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
  <value>org.apache.hadoop.mapred.ShuffleHandler</value>
</property>
<property>
  <name>yarn.resourcemanager.hostname</name>
  <value>127.0.0.1</value>
</property>
<property>
  <name>yarn.acl.enable</name>
  <value>0</value>
</property>
<property>
  <name>yarn.nodemanager.env-whitelist</name>
  <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PERPEND_DISTCACHE,HADOOP_YARN_HOME,HADOOP
</property>
<!-- configuracion de HIVE-->
<property>
  <name>hadoop.proxyuser.db_user.groups</name>
  <value>*</value>
</property>
<property>
  <name>hadoop.proxyuser.db_user.hosts</name>

```

Al presionar “Ctrl + X” aparecerá “Save modified buffer?” escribe “Y”, además por último “Enter”.

Así quedaría ambos códigos de YARN y HIVE

Nota: Estas configuraciones son estándar para un cluster de un solo nodo, según la [guía oficial de Hadoop](#).



```

hdoop@haddop-VirtualBox: ~
hdoop@haddop-VirtualBox:~$ nano $HADOOP_HOME/etc/hadoop/core-site.xml
hdoop@haddop-VirtualBox:~$ nano $HADOOP_HOME/etc/hadoop/hdfs-site.xml
hdoop@haddop-VirtualBox:~$ nano $HADOOP_HOME/etc/hadoop/mapred-site.xml
hdoop@haddop-VirtualBox:~$ nano $HADOOP_HOME/etc/hadoop/yarn-site.xml
hdoop@haddop-VirtualBox:~$

```

Paso 11: Formateo del Namenode

El Namenode gestiona el sistema de archivos HDFS y debe formatearse antes de su primer uso.

1. Ejecuta: `hdfs namenode -format`

```
hadoop@hadoop-VirtualBox: ~  
hadoop@hadoop-VirtualBox:~$ hdfs namenode -format  
WARNING: /home/hadoop/hadoop-3.4.1/logs does not exist. Creating.  
2025-04-23 19:43:20,243 INFO namenode.NameNode: STARTUP_MSG:  
/*****  
STARTUP_MSG: Starting NameNode  
STARTUP_MSG: host = hadoop-VirtualBox/127.0.1.1  
STARTUP_MSG: args = [-format]  
STARTUP_MSG: version = 3.4.1  
STARTUP_MSG: classpath = /home/hadoop/hadoop-3.4.1/etc/hadoop:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/slf4j-ap  
t-1.7.36.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/stax2-api-4.2.1.jar:/home/hadoop/hadoop-3.4.1/share/hadoop  
common/lib/netty-resolver-dns-4.1.100.Final.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/commons-math3-3.6.1.jar  
:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/netty-transport-classes-epoll-4.1.100.Final.jar:/home/hadoop/hadoop-3.4  
.1/share/hadoop/common/lib/snappy-java-1.1.10.4.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/j2objc-annotations-  
1.1.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/commons-beanutils-1.9.4.jar:/home/hadoop/hadoop-3.4.1/share/hado  
op/common/lib/jersey-core-1.19.4.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/reload4j-1.2.22.jar:/home/hadoop/ha  
doo-3.4.1/share/hadoop/common/lib/checker-qual-2.5.2.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/failureaccess  
-1.0.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/curator-client-5.2.0.jar:/home/hadoop/hadoop-3.4.1/share/hadoop  
common/lib/re2j-1.1.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/curator-framework-5.2.0.jar:/home/hadoop/hadoop  
-3.4.1/share/hadoop/common/lib/netty-transport-rxtx-4.1.100.Final.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/k  
erb-crypto-2.0.3.jar:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/kerb-util-2.0.3.jar:/home/hadoop/hadoop-3.4.1/share
```

2. Confirma el formateo cuando se solicite.

```
2025-04-23 19:56:50,616 INFO namenode.FSImageFormatProtobuf: Saving image file /home/hadoop/tmpdata/dfs/name/current/fsim  
age.ckpt 00000000000000000000 using no compression  
2025-04-23 19:56:50,757 INFO namenode.FSImageFormatProtobuf: Image file /home/hadoop/tmpdata/dfs/name/current/fsimage.ckp  
t 00000000000000000000 of size 400 bytes saved in 0 seconds .  
2025-04-23 19:56:50,777 INFO namenode.NNStorageRetentionManager: Going to retain 1 images with txid >= 0  
2025-04-23 19:56:50,784 INFO blockmanagement.DatanodeManager: Slow peers collection thread shutdown  
2025-04-23 19:56:50,815 INFO namenode.FSNamesystem: Stopping services started for active state  
2025-04-23 19:56:50,815 INFO namenode.FSNamesystem: Stopping services started for standby state  
2025-04-23 19:56:50,828 INFO namenode.FSImage: FSImageSaver clean checkpoint: txid=0 when meet shutdown.  
2025-04-23 19:56:50,828 INFO namenode.NameNode: SHUTDOWN_MSG:  
/*****  
SHUTDOWN_MSG: Shutting down NameNode at hadoop-VirtualBox/127.0.1.1  
*****/  
hadoop@hadoop-VirtualBox:~$
```

Si tenemos este mensaje significa que el formateo estuvo correcto.

Paso 12: Inicio de Servicios de Hadoop

1. Inicia los servicios de HDFS y YARN: `./start-dfs.sh; ./start-yarn.sh`

2. Verifica que los servicios estén corriendo: `jps`

- Deberías ver procesos como NameNode, DataNode, ResourceManager y NodeManager.

```
/*****  
SHUTDOWN_MSG: Shutting down NameNode at hadoop-VirtualBox/127.0.1.1  
*****/  
hadoop@hadoop-VirtualBox:~$ start-yarn.sh; start-dfs.sh  
Starting resourcemanager  
Starting nodemanagers  
Starting namenodes on [localhost]  
Starting datanodes  
Starting secondary namenodes [hadoop-VirtualBox]  
2025-04-23 20:24:34,073 WARN util.NativeCodeLoader: Unable to load native-hadoop  
library for your platform... using builtin-java classes where applicable  
hadoop@hadoop-VirtualBox:~$ jps  
2817 ResourceManager  
3047 NodeManager  
4104 Jps  
3977 SecondaryNameNode  
3580 NameNode  
hadoop@hadoop-VirtualBox:~$
```

Paso 13: Instalación de Hive

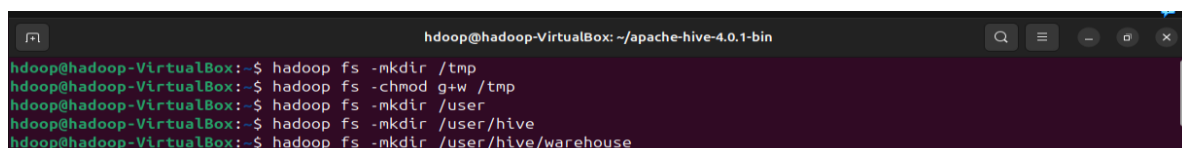
Hive es una herramienta que permite realizar consultas SQL-like sobre datos almacenados en HDFS.

1. Descarga Hive 4.0.1: **wget https://downloads.apache.org/hive/hive-4.0.1/apache-hive-4.0.1-bin.tar.gz**
2. Extrae el archivo: **tar xzf apache-hive-4.0.1-bin.tar.gz**



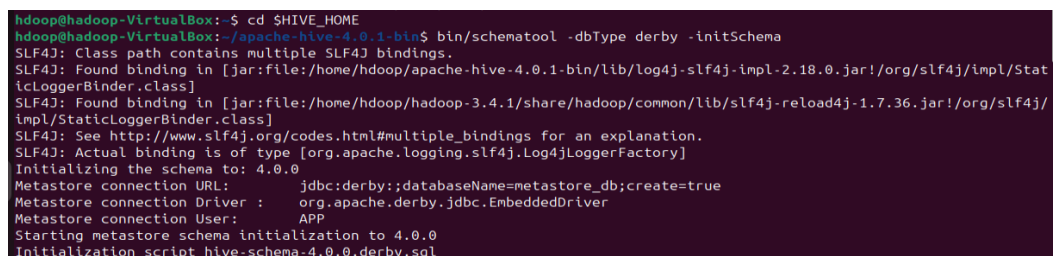
```
hadoop@hadoop-VirtualBox: ~  
hadoop@hadoop-VirtualBox:~$ wget http://downloads.apache.org/hive/hive-4.0.1/apache-hive-4.0.1-bin.tar.gz  
--2025-04-23 20:31:05-- http://downloads.apache.org/hive/hive-4.0.1/apache-hive-4.0.1-bin.tar.gz  
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f8:10a:39da::2, ...  
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:80... connected.  
HTTP request sent, awaiting response... 301 Moved Permanently  
Location: https://downloads.apache.org/hive/hive-4.0.1/apache-hive-4.0.1-bin.tar.gz [following]  
--2025-04-23 20:31:05-- https://downloads.apache.org/hive/hive-4.0.1/apache-hive-4.0.1-bin.tar.gz  
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 408225272 (389M) [application/x-gzip]  
Saving to: 'apache-hive-4.0.1-bin.tar.gz.1'  
  
apache-hive-4.0.1-bin.tar.gz.1 100%[=====] 389.31M 4.93MB/s in 61s  
  
2025-04-23 20:32:07 (6.38 MB/s) - 'apache-hive-4.0.1-bin.tar.gz.1' saved [408225272/408225272]  
  
hadoop@hadoop-VirtualBox:~$ tar xzf apache-hive-4.0.1-bin.tar.gz  
hadoop@hadoop-VirtualBox:~$ ls  
apache-hive-4.0.1-bin  apache-hive-4.0.1-bin.tar.gz.1  hadoop-3.4.1  snap  
apache-hive-4.0.1-bin.tar.gz  dfsdata  hadoop-3.4.1.tar.gz  tmpdata  
hadoop@hadoop-VirtualBox:~$
```

3. Crea directorios en HDFS para Hive:
 - o `hadoop fs -mkdir /tmp`
 - o `hadoop fs -chmod g+w /tmp`
 - o `hadoop fs -mkdir -p /user/`
 - o `hadoop fs -mkdir -p /user/hive`
 - o `hadoop fs -mkdir -p /user/hive/warehouse`



```
hadoop@hadoop-VirtualBox: ~/apache-hive-4.0.1-bin  
hadoop@hadoop-VirtualBox:~$ hadoop fs -mkdir /tmp  
hadoop@hadoop-VirtualBox:~$ hadoop fs -chmod g+w /tmp  
hadoop@hadoop-VirtualBox:~$ hadoop fs -mkdir /user  
hadoop@hadoop-VirtualBox:~$ hadoop fs -mkdir /user/hive  
hadoop@hadoop-VirtualBox:~$ hadoop fs -mkdir /user/hive/warehouse
```

4. Inicializa el metastore de Hive: **schematool -initSchema -dbType derby**



```
hadoop@hadoop-VirtualBox:~$ cd $HIVE_HOME  
hadoop@hadoop-VirtualBox:~/apache-hive-4.0.1-bin$ bin/schematool -dbType derby -initSchema  
SLF4J: Class path contains multiple SLF4J bindings.  
SLF4J: Found binding in [jar:file:/home/hadoop/apache-hive-4.0.1-bin/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop-3.4.1/share/hadoop/common/lib/slf4j-reload4j-1.7.36.jar!/org/slf4j/impl/StaticLoggerBinder.class]  
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.  
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]  
Initializing the schema to: 4.0.0  
Metastore connection URL: jdbc:derby:;databaseName=metastore_db;create=true  
Metastore connection Driver : org.apache.derby.jdbc.EmbeddedDriver  
Metastore connection User: APP  
Starting metastore schema initialization to 4.0.0  
Initialization script hive-schema-4.0.0.derby.sql
```

Paso 14: Configuración Opcional de Spark

Spark es una plataforma para procesamiento rápido de datos, integrable con Hadoop.

1. Regresamos a home: **cd**
2. Descarga Spark (por ejemplo, 3.5.3): **wget**
https://dlcdn.apache.org/spark/spark-3.5.3/spark-3.5.3-bin-hadoop3.tgz
3. Extrae el archivo: **tar xzf spark-3.5.3-bin-hadoop3.tgz**

```
Initialization script completed
hadoop@hadoop-VirtualBox:~/apache-hive-4.0.1-bin$ cd
hadoop@hadoop-VirtualBox:~$ wget https://downloads.apache.org/spark/spark-3.5.5/spark-3.5.5-bin-hadoop3.tgz
--2025-04-23 20:46:36-- https://downloads.apache.org/spark/spark-3.5.5/spark-3.5.5-bin-hadoop3.tgz
Resolving downloads.apache.org (downloads.apache.org)... 135.181.214.104, 88.99.208.237, 2a01:4f8:10a:39da::2, ...
Connecting to downloads.apache.org (downloads.apache.org)|135.181.214.104|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 400724056 (382M) [application/x-gzip]
Saving to: 'spark-3.5.5-bin-hadoop3.tgz'

spark-3.5.5-bin-hadoop3.tgz  100%[=====] 382.16M  6.95MB/s   in 75s

2025-04-23 20:47:51 (5.13 MB/s) - 'spark-3.5.5-bin-hadoop3.tgz' saved [400724056/400724056]

hadoop@hadoop-VirtualBox:~$ tar xzf spark-3.5.5-bin-hadoop3.tgz
hadoop@hadoop-VirtualBox:~$ ls
apache-hive-4.0.1-bin      tmpdata
apache-hive-4.0.1-bin.tar.gz  hadoop-3.4.1
apache-hive-4.0.1-bin.tar.gz.1  hadoop-3.4.1.tar.gz
hadoop@hadoop-VirtualBox:~$ spark-3.5.5-bin-hadoop3
spark-3.5.5-bin-hadoop3.tgz
```

Nota: La configuración de Spark es opcional y depende de tus necesidades. Consulta la [documentación de Spark](#) para más detalles.

Verificación del Entorno

Para asegurarte de que todo está configurado correctamente:

1. **Hadoop:** Accede a la interfaz web en **http://localhost:9870** para HDFS y **http://localhost:8088** para YARN.
2. **Hive:** Ejecuta hive en la terminal y prueba una consulta simple como SHOW DATABASES;.
3. **Spark:** Ejecuta spark-shell para verificar que Spark está funcional.

Tabla de Resumen de Configuraciones

Componente	Versión	Ruta de Instalación	Configuraciones Clave
Java	OpenJDK 11	/usr/lib/jvm/java-11-openjdk-amd64	JAVA_HOME en .bashrc y hadoop-env.sh

Componente	Versión	Ruta de Instalación	Configuraciones Clave
Hadoop	3.4.1	/home/hadoop/hadoop-3.4.1	core-site.xml, hdfs-site.xml, mapred-site.xml, yarn-site.xml
Hive	4.0.1	/home/hadoop/apache-hive-4.0.1-bin	HIVE_HOME, directorios en HDFS
Spark	3.5.3	/home/hadoop/spark-3.5.3-bin-hadoop3	SPARK_HOME en .bashrc

Consejos para el Aprendizaje

- **Entender los Componentes:** Familiarízate con los roles de HDFS (almacenamiento), YARN (gestión de recursos) y MapReduce (procesamiento) en Hadoop.
- **Practicar Comandos:** Usa comandos como `hadoop fs -ls` para explorar HDFS y `hive` para consultas SQL.
- **Consultar Documentación:** Las guías oficiales de [Hadoop](#), [Hive](#), y [Spark](#) son recursos valiosos.
- **Experimentar:** Crea pequeños trabajos MapReduce o consultas Hive para entender cómo funcionan.

Solución de Problemas Comunes

- **SSH no conecta:** Verifica los permisos de `~/.ssh/authorized_keys` y asegúrate de que el servicio SSH esté activo.
- **Hadoop no inicia:** Revisa los logs en `$HADOOP_HOME/logs` y confirma que `JAVA_HOME` esté correctamente configurado.
- **Errores de Hive:** Asegúrate de que los directorios en HDFS estén creados y que el metastore esté inicializado.

Conclusión

Este manual te permite configurar un entorno completo de Hadoop en una máquina virtual, listo para aprendizaje y experimentación con Big Data. Incluye todos los pasos del archivo proporcionado, con correcciones (como el uso de Java 11) y configuraciones estándar para un cluster de un solo nodo. Las explicaciones adicionales y recursos facilitan la comprensión para principiantes.

Key Citations

- [Apache Hadoop 3.4.1 Single Node Cluster Setup](#)
- [Running Hadoop on Ubuntu Linux Single-Node Cluster](#)